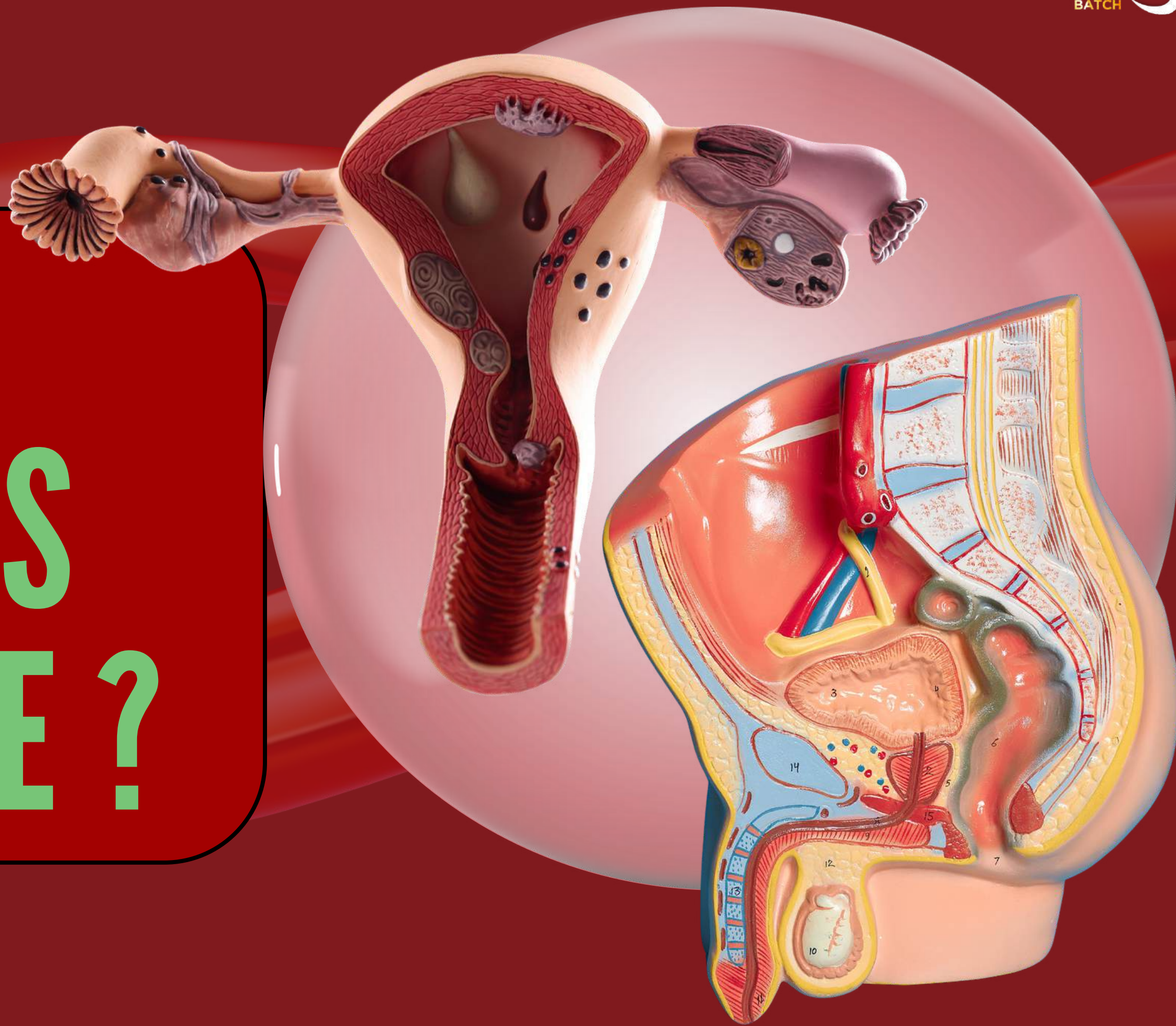


L5

CLASS X - SCIENCE

HOW DO ORGANISMS REPRODUCE ?

PRASHANT KIRAD





Josh Meter

REPRODUCTIVE HEALTH



Reproductive health

- *It is the overall well-being of an individual in all aspects related to reproduction, including physical, emotional, and social health.*
- *It involves safe reproductive practices, awareness, and the prevention of diseases.*

Sexual maturation



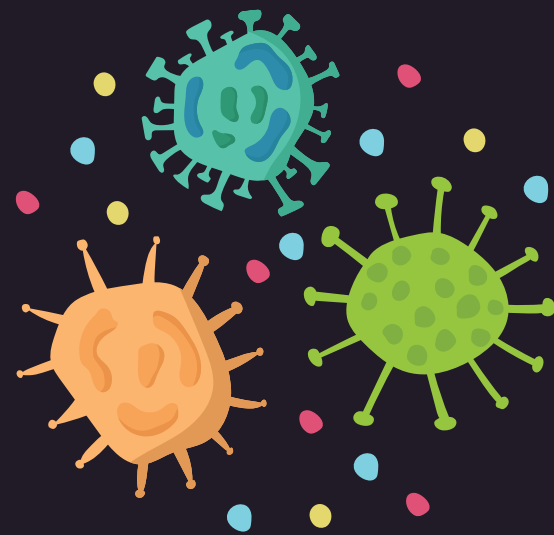
Physical and mental readiness for sexual acts or raising children

Sexual maturation is gradual and occurs while general body growth is still continuing.

Why is reproductive health important?

Reproductive health involves understanding the physical, mental, emotional and social aspects associated with reproduction.

Prevents from reproductive diseases and infections.



Ensures healthy pregnancy and child-birth.



Create awareness about safe and responsible reproductive practice



Why should avoid pregnancy?

To control
population
growth.

To maintain a
proper gap between
two children.



Therefore, methods have been developed to prevent pregnancy.



Sexually Transmitted Diseases (STDs):

Sexually Transmitted Diseases are infections that spread from one person to another mainly through sexual contact. Examples include AIDS, gonorrhea, and syphilis.

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graph TD; STDs[Sexually Transmitted Diseases] --- Bacterial[Bacterial Infections]; STDs --- Viral[Viral infections]; Bacterial --- BacterialList[Gonorrhea, Syphilis]; Viral --- ViralList[HIV-AIDS, Genital Warts]
```

Bacterial Infections

Gonorrhea, Syphilis

Viral infections

HIV-AIDS, Genital Warts

CONTRACEPTION

Contraception is the method of preventing pregnancy by stopping the sperm from reaching the egg, or by preventing fertilisation and implantation.



Methods of Contraception

Mechanical/barrier method

- Condom
- Cervical cap



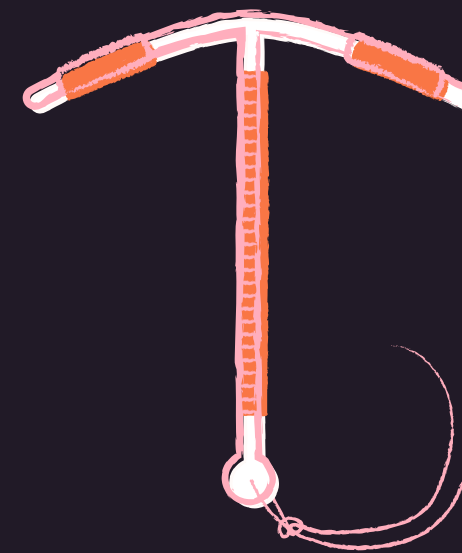
Chemical methods

- Pills



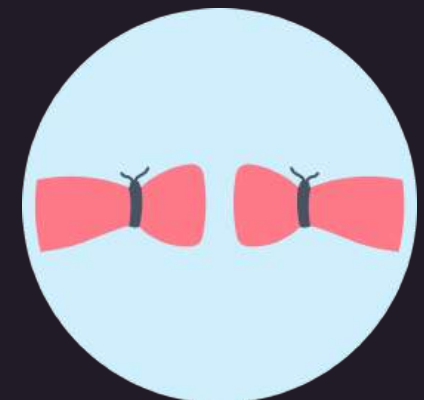
Intra Uterine Contraceptive devices (IUCD)

- Loop
- Copper -T



Surgical Method

- Vasectomy
- Tubectomy



Mechanical/ Physical barrier

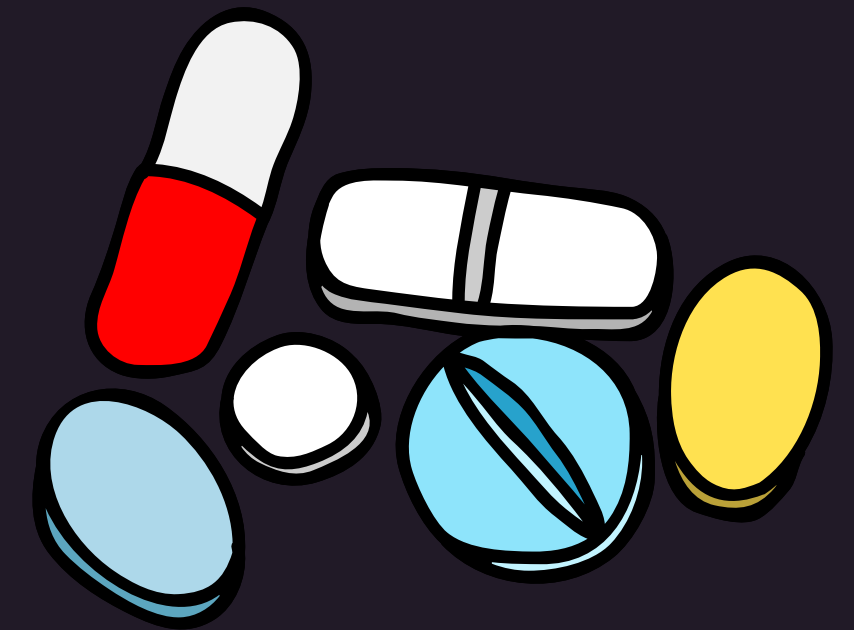
Q. What are the different methods of contraception? Give one example of each.

- Barrier methods, such as **condoms** (for males and females) and diaphragms (for females), prevent the sperm and ovum from meeting.
- They also help prevent the transmission of sexually transmitted diseases by blocking the intermixing of body fluids.



Chemical methods

- Hormonal contraceptives, such as **oral and vaginal pills**, prevent ovulation, alter cervical mucus, or change the uterine lining to prevent implantation.
- These methods may have side effects and do not protect against sexually transmitted diseases (STDs).



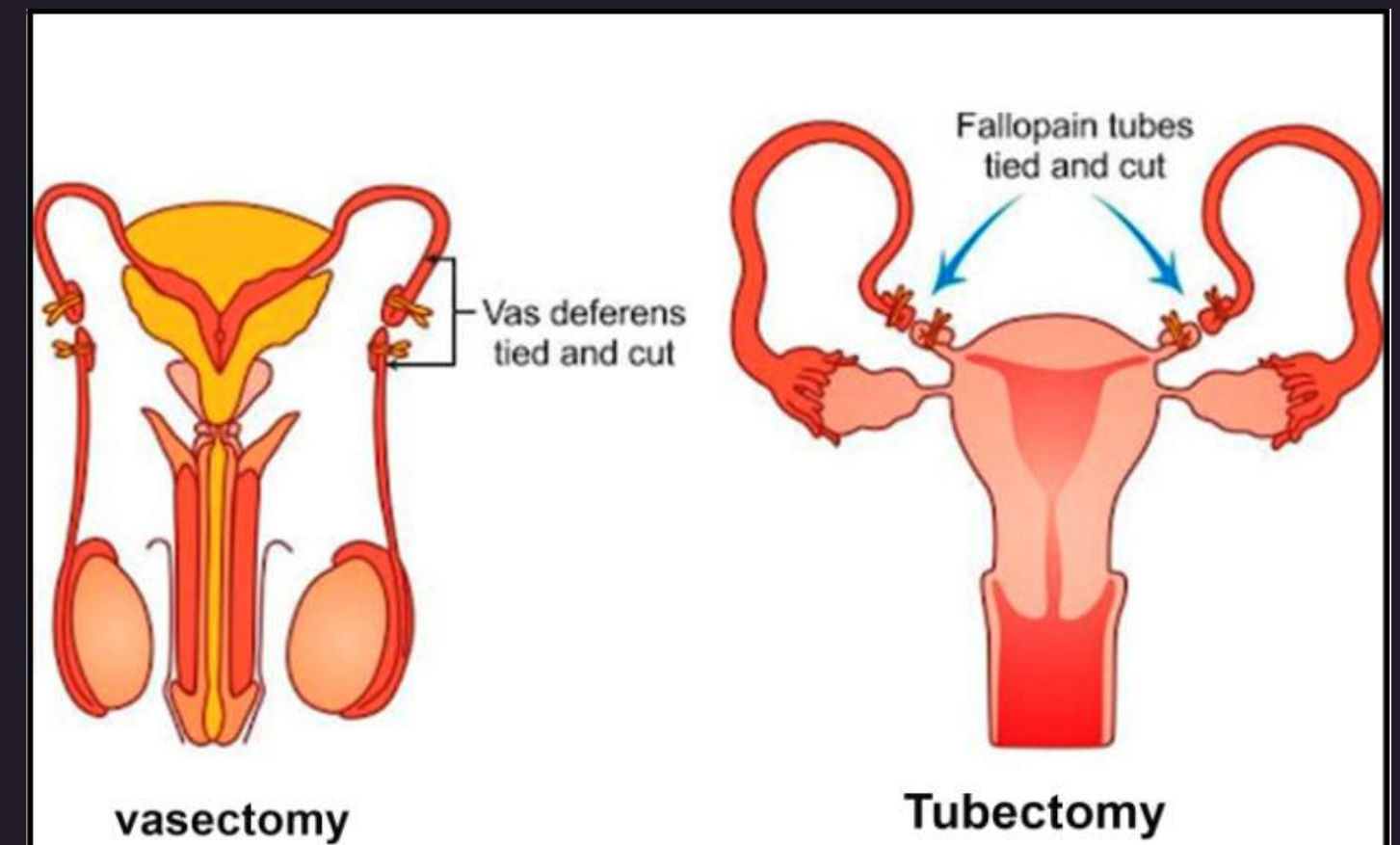
Intra Uterine Contraceptive devices (IUCD)

- These are small T-shaped devices that are planted in the uterus.
- **Copper-T or Loop** placed in the uterus to prevent pregnancy.
- Can cause uterine irritation in some cases.



Surgical Method

- **Male (Vasectomy):** Blocking the vas deferens to prevent sperm transfer.
- **Female (Tubectomy):** Blocking the fallopian tubes to prevent egg transfer.
- Surgery is safe in the long run but may have risks like infection.



Female Foeticide

- Sex-selective abortion of female foetuses is **illegal**.
- Such practices **disturb the female–male sex ratio** in society.
- A balanced sex ratio is important for a healthy society.
- Prenatal sex determination has been prohibited by law.
- Misuse of medical procedures for sex-selective abortion must be prevented



Check yourself

Q. Which among the following diseases is not sexually transmitted?

- a) Syphilis**
- b) Hepatitis**
- c) HIV-AIDS**
- d) Gonorrhea**

Check yourself

Q. Assertion (A): Barrier methods of contraception are preferred for preventing STDs.

Reason (R): Barrier methods chemically kill pathogens during sexual contact.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).**
- (b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).**
- (c) (A) is true but (R) is false.**
- (d) (A) is false but (R) is true.**

Check yourself

Q. Assertion (A): Oral contraceptive pills and copper-T do not prevent sexually transmitted diseases.

Reason (R): Sexually transmitted diseases are transmitted by contact with mucous membranes of infected organs.

- A. Both A and R are true and R is the correct explanation of A.**
- B. Both A and R are true but R is not the correct explanation of A.**
- C. A is true but R is false.**
- D. A is false but R is true.**

Check yourself

Q. “Use of a condom is beneficial for both sexes involved in a sexual act.” Justify this statement by giving two reasons.

(ii) How does an oral contraceptive help in avoiding pregnancies? What is sex-selective abortion?

(iii) How does it affect a healthy society? State any one consequence.

(i) Justification: Use of condom is beneficial for both sexes

1. It prevents unwanted pregnancies.

2. It protects both partners from sexually transmitted diseases (STDs) such as HIV/AIDS.

(ii) Oral contraceptive pills contain hormones that prevent ovulation and make the uterine lining unsuitable for implantation.

Sex-selective abortion: It is the practice of terminating a pregnancy based on the sex of the foetus.

(iii) Sex-selective abortion leads to an imbalanced sex ratio, which affects the healthy structure of society and causes social problems.

Check yourself

Q. Q. (a) What is sex-selective abortion?

(b) Why is prenatal sex determination prohibited by law?

(c) How can sex-selective abortion affect the female–male sex ratio and society?

(a) Sex-selective abortion is the termination of a pregnancy on the basis of the sex of the foetus, especially to avoid the birth of a female child.

(b) It is prohibited to prevent misuse of medical procedures for sex-selective abortion.

(c) It can disturb the female–male sex ratio and create an imbalance in society. A balanced sex ratio is important for a healthy society.

Check yourself

Q. A young couple, Riya and Arjun, visit a gynecologist for family planning advice. They express that they want to delay pregnancy for a few years but are concerned about side effects of medicines and want to avoid STDs. The doctor explains various methods and their effectiveness.

(A) Which contraceptive method should the doctor recommend to prevent both pregnancy and STDs?

(B) If the couple opts for oral pills instead, explain how this method works.

OR

(B) Why is the copper-T not suitable for their primary concern?

(C) State one major advantage of using barrier methods over hormonal methods

Answer: (A) The doctor should recommend barrier methods (condoms) because they prevent both pregnancy and STDs.

(B) Oral contraceptive pills contain hormones that:

- Prevent ovulation (release of egg).
- Thicken cervical mucus to block sperm entry.
- Make the uterine lining unsuitable for implantation.

OR

(B) Copper-T prevents pregnancy but does not protect against STDs, which is their major concern.

(C) Barrier methods do not cause hormonal side effects and also protect against STDs.

Check yourself

Q: Name the part/organ of the human female reproductive system

1. where contraceptive devices such as loops or copper Ts are placed to prevent pregnancy.
2. which is blocked to prevent the transfer of eggs.
3. where the formation of green cells as ova takes place.
4. from where the embryo gets nutrition from the mother's blood.

- Uterus
- Fallopian tube (Oviduct)
- Ovary
- Placenta

Check yourself

Q: Differentiate between vasectomy and tubectomy. How do these methods prevent fertilisation?

- **Vasectomy:** A surgical method in males in which the vas deferens is blocked, so sperms cannot be transported.
- **Tubectomy:** A surgical method in females in which the fallopian tubes are blocked, so the egg cannot meet the sperm.
- **How they prevent fertilisation:** In both methods, the sperm and egg are prevented from meeting, so fertilisation does not occur.

Check yourself

Q: State the possible side effects of oral pills, Copper-T and surgical contraceptive methods.

- Oral pills: May cause side effects because they change the hormonal balance of the body.
- Copper-T: May cause irritation in the uterus.
- Surgical methods: Surgery may cause infections and other complications if not performed properly.

Check yourself

Q: What are the different methods of contraception used to prevent unwanted pregnancy? Explain any three methods. Which contraceptive method also helps in reducing the transmission of sexually transmitted diseases? Name one bacterial and one viral STD. (5 Marks)

Answer

- **Barrier method – Condoms:** Prevent sperm from reaching the egg. Condoms also help reduce transmission of many STDs.
- **Hormonal method – Oral pills:** Change the hormonal balance so that eggs are not released.
- **IUCD – Copper-T/Loop:** Placed inside the uterus to prevent pregnancy.
- **Surgical methods:**
 - Vasectomy:** Vas deferens is blocked in males.
 - Tubectomy:** Fallopian tubes are blocked in females.
- **Examples of STDs:**
 - Bacterial:** Gonorrhoea / Syphilis
 - Viral:** Warts / HIV-AIDS

